DOCUMENT C | SOLUTIONS 3

Worked solutions and proposed marking scheme

Challenging application-based practice paper. These are independent suggested answers and mark allocations, not official university solutions.

Award method marks for correct reasoning and units even when a later arithmetic slip occurs; avoid penalizing the same carried error repeatedly. Equivalent physically correct explanations and valid stated assumptions can earn credit. A part's marking points total its question allocation. Diagrams are schematic and not to scale. The complete question text is repeated below so each solution file is usable independently.

Source notation

S1-S06 refer to the supplied six lecture decks and PDF page/slide numbers. A1/A2 are the supplied thermal/OCT student assignments, cited by PDF page including covers. "Recurring family" identifies the historical concept, not copied wording; original extensions are labeled. A historical reference means the closest related question, not evidence that the exact model task appeared.

Equation corrections

Positive activity is -dN/dt. Conventional spin echoes use intrinsic T2 in the ideal echo envelope, whereas gradient echoes retain T2* sensitivity. RF bandwidth means full width unless explicitly stated otherwise. All calculations use γ=267.52×10^6 rad s^-1 T^-1 and unrounded intermediate values.

Model 3 | Question 1: Image quality and quantitative nuclear imaging

1(a) [4 marks]

Question. Two systems have normalized Gaussian PSFs. System A has σx=0.40 mm and σy=1.00 mm; B has σx=σy=0.70 mm. Calculate each directional FWHM and choose the better system for separating structures 1.2 mm apart along x, using separation greater than FWHM as the stated criterion.

Expected answer and working. FWHM=2.35482σ. For A, widths are 0.94193 mm in x and 2.35482 mm in y. For B, both widths are 1.64837 mm. At 1.2 mm separation along x, A satisfies the stated criterion while B does not. A is anisotropic and is worse than B in y, so it is not universally superior; orientation matters.

Proposed allocation: A widths (1); B widths (1); A chosen under stated criterion (1); Directional limitation (1). **Total 4.**

Source map: S1 pp.39-48. **Closest past question:** 2024 Q1(a)ii-iii; 2025 Q1(a). **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(b) [4 marks]

Question. Two regions have mean signals 120 and 100 units and noise standard deviation 5 units. Compute their CNR. A small lesion voxel contains 25% of the first region and 75% of the second. Find its mean signal and CNR against the second region, assuming unchanged noise.

Expected answer and working. For pure regions, CNR=(120-100)/5=4. The mixed voxel signal is 0.25·120+0.75·100=105 units. Against the 100-unit background its CNR is (105-100)/5=1. Partial volume reduces apparent contrast even when receiver noise is unchanged, so detectability deteriorates.

Proposed allocation: CNR=4 (1); Mixture formula (1); 105 units (1); CNR=1 with partial-volume explanation (1). **Total 4.**

Source map: S1 pp.55-56. **Closest past question:** 2024 Q2(a)iii is closest contrast precedent. **Status:** Original syllabus application.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(c) [2 marks]

Question. An image with independent zero-mean noise has SNR=12. How many total independent acquisitions must be averaged to reach SNR=30? Explain why identical-view ultrasound speckle need not obey this scaling.

Expected answer and working. SNRN=SNR1√N gives N=(30/12)²=6.25. At least 7 whole acquisitions are needed (six would give only 29.39). Identical-view speckle is coherent and correlated, so its pattern does not average away as independent noise would.

Proposed allocation: Seven total acquisitions from square law (1); Speckle correlation explanation (1). **Total 2.**

Source map: S1 pp.52-53; S05 pp.32-35. **Closest past question:** 2024 Q4(a)iii is closest compounding precedent. **Status:** Original syllabus application.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(d) [6 marks]

Question. A tracer has physical half-life 8 h and biological half-life 12 h. Its initial in-body activity is 160 MBq. Calculate the effective half-life, activity after 6 h, and time at which activity first falls to 20 MBq. Explain the role of independent clearance.

Expected answer and working. λeff=λphys+λbio, hence 1/Teff=1/8+1/12=5/24 h^-1 and Teff=4.8 h. After 6 h, A=160·2^(-6/4.8)=67.27171 MBq. Reaching 20 MBq is a factor-of-eight decrease, or three effective half-lives: t=3·4.8=14.4 h. Independence and exponential clearance allow the survival factors to multiply and the rate constants to add; effective half-life is shorter than either component.

Proposed allocation: Sum-rate law and assumption (1); 4.8 h (1); 6 h decay substitution (1); 67.27 MBq (1); Three half-lives (1); 14.4 h with interpretation (1). **Total 6.**

Source map: S06 pp.6-8. **Closest past question:** 2024 Q1(b)iv. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(e) [5 marks]

Question. A non-paralyzable gamma-camera channel records n=40,000 counts/s. Its dead time is τ=5.0 μs. Using τ=(N-n)/(nN), derive N in terms of n and τ, calculate the true rate N and the fraction of true events lost.

Expected answer and working. Rearrange nNτ=N-n to N(1-nτ)=n, hence N=n/(1-nτ). Here nτ=40000·5e-6=0.20, so N=40000/0.80=50,000 counts/s. Fraction lost=(N-n)/N=10000/50000=0.20, or 20%. The formula requires nτ<1; its measured-rate saturation limit is 1/τ=200,000 counts/s. A paralyzable model would not use this same correction.

Proposed allocation: Algebraic rearrangement (2); Unit substitution and N=50,000 (1); Lost fraction=20% (1); Validity / model interpretation (1). **Total 5.**

Source map: S06 pp.14-17. **Closest past question:** No direct historical dead-time question. **Status:** Original syllabus application.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(f) [4 marks]

Question. Describe true, random and scattered coincidences in PET. Explain one way that false coincidences affect the reconstructed image.

Expected answer and working. A true coincidence records an unscattered pair from the same annihilation, defining the appropriate LOR. A random coincidence records photons from different annihilations inside the acceptance window. A scattered coincidence includes a photon deflected before detection, misplacing the inferred LOR even if it belongs to the same annihilation. Random/scattered events add incorrectly located counts, lowering contrast and quantitative accuracy unless corrected.

Proposed allocation: True definition (1); Random definition (1); Scattered definition (1); Image consequence (1). **Total 4.**

Source map: S06 pp.32-34. **Closest past question:** 2025 Q1(b)ii is closest coincidence precedent. **Status:** Original syllabus application.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

Model 3 | Question 2: Radiographic design and reconstruction constraints

2(a) [4 marks]

Question. An X-ray target has actual projected electron-track width F=2.0 mm and bevel angle θ=12°. Calculate effective focal spot f=F sinθ and beam coverage 2D tanθ at D=80 cm. Recalculate at 8° and explain the image/FOV trade-off.

Expected answer and working. At 12°, f=2 sin12°=0.41582 mm; coverage=160 tan12°=34.0090 cm. At 8°, f=0.27835 mm and coverage=22.4865 cm. The smaller bevel angle reduces effective spot and geometric blur but narrows the field available at the same source distance. These projected-spot calculations hold F constant and do not themselves change the actual electron-track heating area.

Proposed allocation: 12° spot and coverage (1); 8° spot and coverage (1); Improved geometric resolution (1); Reduced coverage / fixed-F interpretation (1). **Total 4.**

Source map: S2 pp.9-13. **Closest past question:** 2025 Q2(a)ii-iii. **Status:** Original numerical variant.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(b) [4 marks]

Question. Without a grid, a detector receives 600 primary and 400 scattered photons in a stated interval. A grid transmits 70% of primary photons and 10% of scattered photons. Find the detected total and primary fraction with the grid. If output is scaled to restore the original total detected count, estimate the relative exposure factor; assume counts scale linearly with exposure.

Expected answer and working. Grid counts are P=0.70·600=420 and S=0.10·400=40, total 460. Primary fraction rises from 600/1000=60% to 420/460=91.304%. To restore 1000 total counts requires a factor 1000/460=2.17391 in exposure under the stated linear model. Scatter suppression can improve contrast but maintaining detector counts increases patient exposure. This simplified count-matching factor is not a universal grid Bucky factor.

Proposed allocation: 420 primary + 40 scatter (1); 460 total (1); 91.3% primary (1); 2.174 exposure factor and trade-off (1). **Total 4.**

Source map: S2 pp.34-38. **Closest past question:** 2025 Q2(a)i is closest hardware precedent. **Status:** Original syllabus application.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(c) [5 marks]

Question. A CT scanner measures air as -1040 HU and water as +20 HU. A linear calibration is h=a hm+b. Determine a and b using true air=-1000 HU and water=0 HU. Correct a measured tissue value of 232 HU, then compute its μ if μwater=0.200 cm^-1.

Expected answer and working. The constraints are -1000=-1040a+b and 0=20a+b. Subtracting gives 1000=1060a, so a=50/53=0.94339623 and b=-1000/53=-18.86792453 HU. For hm=232, h=(50·232-1000)/53=200 HU. HU=1000(μ-μwater)/μwater gives μ=μwater(1+h/1000)=0.200·1.2=0.240 cm^-1. The corrected value, not the raw measured value, enters the attenuation conversion.

Proposed allocation: Two calibration equations (1); a (1); b (1); Corrected 200 HU (1); μ=0.240 cm^-1 (1). **Total 5.**

Source map: S3 pp.31-32. **Closest past question:** 2025 Q2(b)ii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(d) [8 marks]

Question. For a 2×2 image [[a,b],[c,d]], unit-weight projections are top row=8, bottom row=12, left column=7 and right column=13. Start from zero and use one equal-residual ART sweep in that order. Then an extra isolated ray gives a=2. Determine the unique image satisfying all data and verify it. Distinguish this final constrained solution from the one-sweep estimate.

Expected answer and working. From zero, top update gives [[4,4],[0,0]], then bottom gives [[4,4],[6,6]]. Left predicted sum=10, residual=-3; subtract 1.5 from a,c to obtain [[2.5,4],[4.5,6]]. Right predicted sum=10, residual=+3; add 1.5 to b,d, yielding [[2.5,5.5],[4.5,7.5]]. This fits the row/column data. The new independent constraint sets a=2. Then b=8-2=6; c=7-2=5; d=12-5=7. Thus the unique constrained image is [[2,6],[5,7]], with right sum 6+7=13. Simply setting a=2 in the one-sweep estimate would break existing sums; the other entries must change too. The direct constrained solution is not claimed to be a single extra ART ray update.

Proposed allocation: Top and bottom intermediate images (2); Left residual and update (1); Right residual and one-sweep estimate (1); Use isolated a=2 (1); b=6,c=5,d=7 (1); Verify all projections (1); Distinguish constrained solution from one-sweep ART (1). **Total 8.**

Source map: S3 pp.26-28. **Closest past question:** 2024 Q2(b)iii. **Status:** Original application of ART.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(e) [4 marks]

Question. Explain why noisy or incomplete projections can make exact ray-sum agreement undesirable or impossible. Give one advantage and one limitation of iterative reconstruction relative to direct analytic reconstruction.

Expected answer and working. Noise makes measured rays mutually inconsistent with a single ideal attenuation image, and incomplete views leave unconstrained object features. Iteration can include a forward model of acquisition and seek an approximate statistically appropriate fit, rather than satisfy every noisy ray exactly. It can accommodate incomplete geometry or detector characteristics. Its computational cost is greater and results depend on model/constraints and stopping; uncontrolled ART can oscillate with inconsistent noisy data. Direct analytic methods are fast but usually assume an ideal, adequately sampled geometry.

Proposed allocation: Noisy inconsistency (1); Incomplete-data ambiguity (1); Iterative modeling advantage (1); Computation/stability limitation vs analytic (1). **Total 4.**

Source map: S3 pp.14,26-29. **Closest past question:** 2025 Q2(c)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

Model 3 | Question 3: MRI signal optimization and acquisition design

3(a) [4 marks]

Question. For a spoiled GE sequence, explain the roles of TR, TE, flip angle α and T2*. State which may be chosen by the operator and which is a tissue/field property.

Expected answer and working. TR is the time between successive excitations of the same slice and controls longitudinal recovery. TE is excitation-to-echo-centre time and controls transverse decay weighting. Flip angle is the RF-induced magnetization rotation and affects excitation and steady-state saturation. T2* describes transverse decay including intrinsic T2 and field inhomogeneity. TR, TE and α are acquisition choices; T2* depends on tissue and the local/static field conditions.

Proposed allocation: TR (1); TE (1); α (1); T2* with classification (1). **Total 4.**

Source map: S4 pp.22,29-38,62,78. **Closest past question:** 2024 Q3(b)iii-v. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(b) [5 marks]

Question. For a fixed-TR, fixed-TE spoiled GE sequence with E=exp(-TR/T1), signal is proportional to (1-E)sinα/(1-E cosα). Derive the maximizing angle for $0<\alpha<\pi/2$. Evaluate it for TR=0.40 s and T1=1.00 s.

Expected answer and working. Treat E as constant. Differentiating the angle-dependent fraction gives $(1-E)[\cos\alpha(1-E\cos\alpha)-E\sin^2\alpha]/(1-E\cos\alpha)^2 = (1-E)(\cos\alpha-E)/(1-E\cos\alpha)^2$. Set the numerator to zero: cosα=E. The derivative changes from positive to negative across this point, so it is a maximum. With E=e^-0.4=0.67032005, αE=acos(E)=47.9082°. The result assumes spoiled transverse magnetization and constant noise/other acquisition settings.

Proposed allocation: Correct quotient differentiation (2); Simplification cosα-E (1); Stationary point and maximum (1); 47.91° (1). **Total 5.**

Source map: S4 p.78. **Closest past question:** 2024 Q3(b)iv-v. **Status:** Original derivation of supplied lecture equation.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(c) [5 marks]

Question. At TR=0.40 s, TE=0.012 s and T1=1.00 s, compare two otherwise identical regions with T2*=0.040 s and 0.080 s. Use the Ernst angle from (b). Calculate their normalized GE signals for ρ=1, their signal ratio, and the percentage increase for the longer-T2* region.

Expected answer and working. At the Ernst angle the recovery/flip factor is √[(1-E)/(1+E)]=√[(1-e^-0.4)/(1+e^-0.4)]=0.44426942. For T2*=0.040 s, S1=0.44426942 exp(-0.012/0.040)=0.32912288. For 0.080 s, S2=0.44426942 exp(-0.012/0.080)=0.38238624. Their ratio S2/S1=exp(0.15)=1.16183424, so the second signal is 16.1834% larger. The same density/T1 does not imply equal GE intensity because transverse decay differs.

Proposed allocation: Common steady-state factor (1); S1 calculation (1); S2 calculation (1); Ratio 1.16183 (1); 16.18% increase and interpretation (1). **Total 5.**

Source map: S4 pp.75,78. **Closest past question:** 2024 Q3(b)v. **Status:** Original multi-tissue variant.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(d) [4 marks]

Question. A 256×128 readout-by-phase image uses TR=0.40 s and three averages. Calculate total scan time. If the phase steps are doubled and averages reduced to one, find the new scan time and explain the change.

Expected answer and working. Original time=128·0.40·3=153.6 s. New time=256·0.40·1=102.4 s. The new/original time ratio is 2/3. Each phase step needs a repetition; reducing three averages to one more than offsets doubling phase steps. The readout dimension is not a separate repetition multiplier under this conventional sequence model.

Proposed allocation: Original formula and time (1); New formula and time (1); Ratio 2/3 (1); Repetition explanation (1). **Total 4.**

Source map: S4 pp.61-62,71,92. **Closest past question:** 2025 Q3(b)iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(e) [4 marks]

Question. Each complete slice sequence occupies 25 ms within TR=0.40 s. Find the maximum ideal interleaved slice count and the scan time for 12 slices using the original settings of (d). Explain the required frequency and storage changes.

Expected answer and working. There are floor(400/25)=16 full slice slots per TR. Twelve slices fit, so each receives one excitation per 0.40 s and all 12 complete the original 128 phase steps and three averages in 153.6 s, neglecting preparation/overhead. Change RF centre frequency for slice position, retain the prescribed bandwidth/gradient for thickness, and store each slice's k-space separately. Multiplying 153.6 by 12 would describe serial rather than interleaved acquisition.

Proposed allocation: 16 slots (1); 12 fit /153.6 s (1); RF centre frequency (1); Separate k-space and same per-slice TR (1). **Total 4.**

Source map: S4 pp.72-74. **Closest past question:** 2024 Q3(b)ii; 2025 Q3(b)iv. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(f) [3 marks]

Question. Under the approximate scaling SNR proportional to voxel volume times √(total ADC-on time), what happens to SNR if voxel volume is halved with unchanged ADC-on time? How much additional acquisition time would restore the original SNR? State a practical cost.

Expected answer and working. Halving voxel volume halves SNR at the same total ADC-on time under the stated model. To recover a factor of two in SNR needs a factor of four in acquisition time because √4=2. The longer scan increases motion risk and reduces temporal resolution; signal averaging cannot fully remove motion or correlated artifacts.

Proposed allocation: SNR halves (1); Fourfold time (1); Practical cost (1). **Total 3.**

Source map: S4 p.92; S1 pp.52-53. **Closest past question:** No direct historical voxel-time trade-off question. **Status:** Original syllabus application.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

Model 3 | Question 4: Ultrasound design and optical ranging

4(a) [6 marks]

Question. A target reflects 50% of incident intensity and lies in tissue with attenuation 0.60 dB/(cm·MHz). At f=3 MHz the total permissible echo-intensity loss is 90 dB, measured relative to launched intensity. Calculate the maximum depth, accounting for reflection and round-trip propagation. Ignore all other losses.

Expected answer and working. Reflection contributes a positive loss Lr=-10 log10(0.50)=3.01030 dB. The propagation budget is 90-3.01030=86.98970 dB. For one-way depth d in cm, Lprop=2·0.60·3·d=3.6d dB. Therefore dmax=86.98970/3.6=24.16381 cm. Check: 2·0.60·3·24.16381+3.01030=90 dB. A pressure-amplitude coefficient would require a different dB conversion; here intensity reflectance is explicitly given.

Proposed allocation: Intensity reflection loss with 10 log (1); 3.0103 dB (1); Remaining propagation budget (1); Round-trip attenuation expression (1); 24.16 cm (1); Check / physical interpretation (1). **Total 6.**

Source map: S05 pp.15-16,28; local discussion PDF p.5. **Closest past question:** 2025 Q4(a)iv is closest depth precedent. **Status:** Original application with reflection loss.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(b) [5 marks]

Question. At 3 MHz and c=1540 m/s, the pulse contains three cycles. The focal distance is 60 mm and aperture width 20 mm. Compute wavelength, axial resolution and focal lateral resolution using the supplied lecture approximations. What would doubling frequency do if all other values remained fixed?

Expected answer and working. λ=c/f=1540/(3e6)=0.000513333 m=0.513333 mm. Axial resolution=Ncλ/2=3·0.513333/2=0.770 mm. F-number=60/20=3; lateral resolution≈λF/L=0.513333·3=1.540 mm. Doubling frequency halves both numerical resolution distances, improving detail, but doubles propagation loss per cm in the given attenuation model and reduces accessible depth.

Proposed allocation: Wavelength (1); Axial 0.770 mm (1); F-number 3 (1); Lateral 1.540 mm (1); Frequency trade-off (1). **Total 5.**

Source map: S05 pp.25-29. **Closest past question:** 2025 Q4(a)iii is closest resolution precedent. **Status:** Original numerical variant.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(c) [4 marks]

Question. A 128-line frame images to 20 cm at c=1540 m/s. Compare maximum frame rates for one transmit focus per line and three sequential transmit foci per line. Assume no parallel reception or timing overhead.

Expected answer and working. One pulse echo requires 0.40/1540=259.740 μs. For 128 lines and one focus, frame time=128·0.40/1540=0.0332468 s, so FR=30.078125 frames/s. Three transmit foci require three firings/line, so time triples to 0.0997403 s and FR=10.02604 frames/s. More transmit focal zones improve focusing coverage at the expense of temporal resolution.

Proposed allocation: Round-trip echo time (1); 30.08 frames/s (1); Threefold time /10.03 frames/s (1); Focusing-time trade-off (1). **Total 4.**

Source map: S05 pp.6-7,19,25-29. **Closest past question:** 2024 Q4(a)v-vi. **Status:** Original application of timing.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(d) [4 marks]

Question. Explain angular and frequency compounding for speckle reduction. State why improved smoothness can conceal a loss of temporal resolution.

Expected answer and working. Angular compounding combines registered views acquired with different beam directions; frequency compounding combines views at distinct frequencies within probe bandwidth. Speckle changes because coherent phase combinations differ, while anatomical structure remains more consistent. Acquiring several views for one displayed composite can reduce the number of independent composite frames per second or introduce temporal averaging/motion blur. A smooth display alone does not prove better spatial or temporal resolution.

Proposed allocation: Angular mechanism (1); Frequency mechanism (1); Decorrelated speckle / consistent anatomy (1); Acquisition-time / motion trade-off (1). **Total 4.**

Source map: S05 pp.32-35. **Closest past question:** 2024 Q4(a)iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(e) [6 marks]

Question. An OCT source has Gaussian spectrum with centre wavelength $\lambda_0=840$ nm and FWHM bandwidth $\Delta\lambda=60$ nm. Using $\delta z_{air}=(2\ln 2/\pi)\lambda_0^2/\Delta\lambda$, find axial resolution in air and in tissue of refractive index $n=1.40$, taking $\delta z_{tissue}=\delta z_{air}/n$. Explain why a broader spectrum improves axial resolution and why this calculation does not determine lateral resolution.

Expected answer and working. Use consistent wavelengths: $\lambda_0^2/\Delta\lambda=840^2/60=11760$ nm. Multiplying by $2\ln 2/\pi=0.44127120$ gives $\delta z_{air}=5189.3495$ nm$=5.18935$ μm. In tissue $\delta z=5.18935/1.40=3.70668$ μm. A broader spectrum has a shorter coherence gate, separating reflections at smaller optical delays. Lateral resolution depends on focusing optics/beam spot rather than this temporal-coherence formula. This is assignment-level OCT coverage; it is not a recurring historical calculation.

Proposed allocation: Formula and consistent units (1); 5.189 μm in air (1); 3.707 μm in tissue (1); Broad bandwidth shortens coherence gate (1); Lateral focusing distinction (1); Physical interpretation / correctly distinguishes optical and geometric depth (1). **Total 6.**

Source map: A2 PDF p.5; syllabus p.1; Aumann technical chapter cited in A2. **Closest past question:** No direct historical OCT-resolution numerical. **Status:** Original assignment-supported coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.